

Shaoyi Lu

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EDUCATION

McMaster University <i>M.S: Master of Engineering, Computing and Software</i>	09/2024 - 08/2026
University of Wisconsin, Madison <i>B.S: Major in Computer Science, Mathematics & Data Science</i>	09/2021 - 05/2024

WORKING EXPERIENCE

Johnson Electric Canada Ltd. Jr. Engineering Specialist (Co-op), Smart Factory & Digitization	Ancaster, Ontario, Canada 09/2025 – 05/2026
- Improved a Detectron2-based object detection and instance segmentation system by analyzing false-pass inspection images, refining defect labels, and applying Gaussian filtering to reduce surface artifacts, improving accuracy by approximately 8.5%. - Built Python-based data pipelines to aggregate vision model outputs and calculate defect-specific metrics using scikit-learn, while developing a Django application with structured JSON schemas for inspection data management and Excel-based reporting.	
Xingtu Xinke Technology Co., Ltd. AI Research Assistant, Wireless Signal Classification	Wuhan, Hubei, China 06/2024 – 08/2024
- Designed a Deep Learning model integrating CNN, LSTM, and Self-Attention for 5G wireless signal modulation classification, achieving 94.1% accuracy and improving performance by 3.57% over old CNN models. - Developed a complete PyTorch training pipeline covering signal normalization, train/test splitting, data augmentation, model training, and evaluation to improve classification robustness.	

PROJECTS

Full-Stack Portfolio Management System GitHub	07/2026
- Developed a full-stack personal portfolio platform using React, JavaScript, and Vite, featuring a protected admin dashboard with Firestore Authentication and Cloud Firestore for dynamic content management. - Built a custom CMS that enables updating profile information, work experience, education, projects, skills, and contact details without modifying source code, with changes synchronized directly to Firestore. - Implemented lightweight analytics tracking for page views, resume clicks, project interactions, traffic sources, devices, and browsers, with automated tests validating CMS workflows and UI behavior.	
Multi-Domain Image Classification System with Model Fusion GitHub	05/2026
- Built a multi-domain image classification pipeline using Python, PyTorch, and Torchvision, training multiple compact specialist CNN models using Google Colab GPU and merging them into a unified classifier. - Implemented local-to-global label mapping, expert routing, feature-level fusion, and source-aware weighting to combine predictions across different label spaces and reduce routing errors. - Achieved 88.73% global test accuracy in the disjoint-label setting and improved overlapping-label validation accuracy from 84.40% to 86.67%.	

RESEARCH EXPERIENCE

Spatiotemporal Analysis for Insecticide Resistance Management	05/2023 – 05/2024
- Processed and analyzed data from over 47,000 field locations (2007–2018) using R, performing exploratory analysis, multiple regression, and spatiotemporal modeling to identify key drivers of insecticide resistance. - Developed predictive models that achieved over 90% accuracy in forecasting pest population trends, supporting data-driven recommendations for resistance management.	
Human Motion Analysis Using Computer Vision	08/2023 – 05/2024
- Developed a Python-based RGB-D human motion analysis pipeline using MediaPipe Pose for 2D/3D key-point detection, skeletal tracking, and spatio-temporal motion feature extraction. - Implemented 3D articulated body modeling with BVH motion capture processing, quaternion-based pose representation, and forward kinematics for joint trajectory analysis and biomechanical assessment.	

SKILLS

- Programming Languages:** Python, Java, JavaScript/TypeScript, C/C++, SQL, R, MATLAB, HTML/CSS
- Frameworks & Libraries:** React, Node.js, Django, Vite, Firebase, PyTorch, Torchvision, Detectron2
- Developer Tools & Data:** Git, Docker, Linux, MySQL, PostgreSQL, Excel, Power BI, Tableau